

Asthma — The Asthma Step Ladder (GINA steps)





## 1. Control banner

### WHAT YOU SEE

A glowing 'CONTROL':  $\leq 2$  day/wk,  $\leq 2$  night/mo,  $\leq 1$  exac/yr, normal activity, optimized FEV'.

### WHAT IT ENCODES

Stepping decisions are driven by symptom control over the last 4 weeks: daytime symptoms  $\leq 2 \times$ /week, NO night waking, reliever use  $\leq 2 \times$ /week, and no activity limitation = well-controlled. 1–2 of these = partly controlled; 3–4 = uncontrolled. Future-risk factors ( $\geq 1$  exacerbation/year, low FEV1, high SABA use, poor adherence) are assessed separately and independently justify stepping up even when current symptoms look acceptable.

### BOARD TRAP

Boards test that 'control' is symptom-based over 4 weeks — a single normal spirometry does not equal control, and good current symptoms with a recent exacerbation still counts as not-controlled because exacerbation risk is its own domain.

## 2. Before stepping up checklist

### WHAT YOU SEE

A sign 'before stepping up: diagnosis, technique, adherence, triggers, comorbidities'.

### WHAT IT ENCODES

Before escalating GINA step, verify the 'big five': correct diagnosis (confirm reversible obstruction/variable airflow, not COPD/VCD/heart failure), inhaler technique (most patients use devices wrong), adherence, ongoing trigger/allergen/smoking exposure, and comorbidities (rhinosinusitis, GERD, obesity, OSA). Fixing these resolves most apparent treatment failure without adding drugs.

### BOARD TRAP

The classic vignette: 'uncontrolled' asthmatic on escalating doses — the answer is to check technique/adherence/diagnosis first, NOT to step up therapy or add a biologic.

## 3. Biologic phenotypes key

### WHAT YOU SEE

A box 'anti-IgE=allergic, anti-IL-5=EOS, dupi=IL-4/13, teze=TSLP'.

### WHAT IT ENCODES

Step-5 biologic choice is phenotype/biomarker driven. Anti-IgE (omalizumab) → allergic asthma with  $\uparrow$ IgE and perennial aeroallergen sensitization. Anti-IL-5/5R (mepolizumab, reslizumab, benralizumab) → eosinophilic asthma (high blood eos, frequent exacerbations). Anti-IL-4R $\alpha$  (dupilumab) blocks IL-4 AND IL-13 → eosinophilic or high-FeNO asthma, also helps nasal polyps/atopic dermatitis. Anti-TSLP (tezepelumab) acts at the top of the type-2 cascade and works across phenotypes including low-eosinophil disease.

### BOARD TRAP

Tezepelumab (anti-TSLP) is the one biologic effective regardless of eosinophil/biomarker status; dupilumab is the cross-over choice when nasal polyps or atopic dermatitis coexist.

## 4. Step 1 — PRN reliever only

### WHAT YOU SEE

The lowest rung of the ladder reading 'STEP 1 — PRN reliever only' — the bottom step of GINA therapy: mildest disease, as-needed low-dose ICS-formoterol only.

### WHAT IT ENCODES

Mildest asthma. GINA Track 1 (preferred) = as-needed low-dose ICS-formoterol taken whenever symptomatic — this delivers an anti-inflammatory dose with every reliever puff and cuts severe exacerbations versus SABA. Track 2 (alternative) = take an ICS whenever a SABA is taken.

### BOARD TRAP

SABA-only is NO LONGER recommended at any step — even mild/intermittent asthma needs ICS-containing therapy because SABA-alone increases exacerbation and death risk.

## 5. Step 2 — low-dose ICS or PRN ICS-formoterol

### WHAT YOU SEE

The second rung 'STEP 2 — low-dose ICS or PRN ICS-formoterol' — climbing one step adds daily low-dose ICS (or as-needed ICS-formoterol) for persistent asthma.

### WHAT IT ENCODES

Persistent asthma. Track 1: as-needed low-dose ICS-formoterol. Track 2: daily low-dose maintenance ICS PLUS as-needed SABA. Daily ICS reduces symptoms, exacerbations, and remodeling; adherence is the main limitation for the daily-ICS option.

### BOARD TRAP

LTRA (montelukast) is only an alternative controller here — it is inferior to ICS and carries a boxed warning for neuropsychiatric effects, so it is not first-line.

## 6. Step 3 — low-dose ICS/formoterol

### WHAT YOU SEE

The middle rung 'STEP 3 — low-dose ICS/formoterol' — higher on the ladder a LABA (formoterol) joins the ICS, used as MART.

### WHAT IT ENCODES

Add a LABA to low-dose ICS. Track 1 preferred = low-dose ICS-formoterol used as MART (Maintenance And Reliever Therapy): the same inhaler is taken daily AND for symptom relief. Formoterol is the chosen LABA because its rapid onset lets it double as a reliever, unlike salmeterol.

### BOARD TRAP

MART only works with formoterol-containing combinations — you cannot use a salmeterol inhaler as a reliever because salmeterol has a slow onset.

## 7. Step 4 — medium-dose ICS/formoterol

### WHAT YOU SEE

The fourth rung 'STEP 4 — medium-dose ICS/formoterol' — stepping up bumps the ICS to medium dose and adds LAMA if still uncontrolled.

### WHAT IT ENCODES

Step up to medium-dose ICS-LABA (ideally medium-dose ICS-formoterol MART). Consider adding LAMA (tiotropium) here if still not controlled, and refer to a specialist for phenotyping. Doubling ICS to high dose gives diminishing returns relative to side effects, so adding a second class is favored.

### BOARD TRAP

Adding a LAMA (tiotropium) is the next step before high-dose ICS or biologics — and tiotropium is added to ICS-LABA, never used as monotherapy in asthma.



## 8. Step 5 — high-dose ICS/LABA + LAMA ± biologic

### WHAT YOU SEE

The top rung 'STEP 5 — high-dose ICS/LABA + LAMA ± biologic' — the highest step: triple therapy plus a phenotype-targeted biologic for severe asthma, OCS as last resort.

### WHAT IT ENCODES

Severe asthma: high-dose ICS-LABA + LAMA (triple therapy) and refer for phenotype-targeted biologic (anti-IgE / anti-IL5 / anti-IL4Rα / anti-TSLP). Add-on azithromycin can reduce exacerbations in some adults. Chronic oral corticosteroids are the absolute last resort.

### BOARD TRAP

Chronic OCS cause systemic toxicity (osteoporosis, diabetes, cataracts, adrenal suppression) — start a biologic to spare steroids rather than committing a severe asthmatic to maintenance prednisone.

## 9. Reliever lane (ICS-formoterol)

### WHAT YOU SEE

A blue 'RELIEVER LANE: ICS-formoterol or ICS-SABA' diagonal.

### WHAT IT ENCODES

The reliever 'lane' runs through every step. In GINA Track 1 the reliever at all steps is as-needed low-dose ICS-formoterol; in Track 2 it is as-needed SABA (with ICS taken concurrently). MART (Track 1) means one ICS-formoterol inhaler serves as both controller and reliever, simplifying the regimen and giving anti-inflammatory dosing at the moment of symptoms.

### BOARD TRAP

Mixing the tracks is the trap: you cannot use a SABA as the reliever for a patient on MART ICS-formoterol — the reliever must match the maintenance strategy.

## 10. Step down after 3 months control

### WHAT YOU SEE

An arrow 'step down if controlled 3 months' with alt high/medium-dose ICS.

### WHAT IT ENCODES

Once asthma is well-controlled for ~3 months, step down to find the lowest effective therapy, typically by reducing ICS dose ~25–50% at a time while keeping some ICS on board. Never stop ICS entirely in persistent asthma. Choose a low-risk time to step down (not during a respiratory infection, pregnancy, or travel) and re-check in 1–3 months.

### BOARD TRAP

Stepping down by removing the ICS and leaving LABA behind is the dangerous error — LABA monotherapy is contraindicated. Always taper ICS dose while retaining ICS, and don't step down during instability.

## 11. Frequent SABA = poor control

### WHAT YOU SEE

A figure clutching a reliever inhaler labeled 'frequent SABA = poor control' — heavy blue-inhaler use is the visual red flag for under-treated inflammation and exacerbation/death risk.

### WHAT IT ENCODES

Using ≥3 SABA canisters per year (or needing reliever >2×/week) flags inadequate controller therapy and is an independent predictor of exacerbations and asthma death. High SABA use marks under-treated underlying inflammation.

### BOARD TRAP

The board answer to a high-SABA-use patient is to optimize/step up the ICS-containing controller — NOT to simply refill the SABA prescription.

## 12. LABA-only never in asthma

### WHAT YOU SEE

A warning sign reading 'LABA-only NEVER in asthma' — the standalone long-acting bronchodilator is the forbidden answer because LABA monotherapy raises asthma deaths.

### WHAT IT ENCODES

LABA monotherapy is contraindicated in asthma. The SMART/SALPACA-era data showed LABA-alone increases asthma-related deaths because it masks worsening inflammation while providing bronchodilation. LABA must always be paired with ICS.

### BOARD TRAP

Any answer offering a standalone LABA (salmeterol/formoterol) for asthma is wrong — the LABA must be combined with an ICS. (Note: LABA/LAMA-without-ICS is also wrong in asthma, even though it's used in COPD.)